

Preyash Jain

Data Science | AI Engineering

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MSc Data Science student at TU Dortmund with hands-on experience building production RAG pipelines and multi-LLM systems, specialising in **AI engineering, backend development, and data-driven applications**, with coursework spanning simulation-based inference, computer vision, and natural language processing.

PROFESSIONAL EXPERIENCE

UG Project Volunteer · Gudsky Research Foundation

Oct 2025 – Mar 2026

Remote

- Built a **GenAI-powered assistant** integrating five LLMs with a RAG pipeline to automate email-heavy workflows, **cutting task completion time by roughly 70%**.

AI & Cloud Intern · IBM SkillsBuild × Edunet

Jul – Aug 2025

Remote

- Completed IBM SkillsBuild's **"Getting Started with Artificial Intelligence"** and **"Journey to Cloud"** courses, building foundational knowledge in AI system design and cloud solution architecture.
- Built and evaluated RAG pipelines using LangChain, Docling, and IBM Granite models with Milvus vector storage, benchmarking output quality with **BLEU and ROUGE metrics**.
- Completed IBM's **"Machine Learning with Python"** course (ML0101EN), covering applied classification, regression, and model evaluation techniques.

Intern Executive – School Profiling · Sankalp Concepts

Oct – Dec 2024

Raipur, India

- Conducted **data analysis** to support school profiling for strategic decision-making, using Excel and Google Sheets to streamline data-collection workflows.

EDUCATION

M.Sc. Data Science · TU Dortmund University

2026 – Present

Dortmund, Germany

B.Tech. Computer Science Engineering · Bhilai Institute of Technology

Nov 2021 – Sep 2025

Raipur, India

Higher Secondary Education (Class XII) · PT RD Tiwari Govt. Eng. Medium School

2021

Raipur, India | 91% overall

SKILLS

Languages	Python, JavaScript, TypeScript, SQL
Frameworks & Libs	LangChain, FastAPI, Flask, React, Next.js
Tools & Platforms	Docker, Git, HuggingFace
Databases	PostgreSQL, Supabase
Data & AI	RAG Pipelines, LLM Integration, NLP, Simulation-Based Inference, Machine Learning

PROJECTS

Gravitational Wave Source Parameter Estimation

SBI Course Group Project

BayesFlow, PyCBC, Python

Built a **simulation-based inference pipeline** (BayesFlow) estimating six binary black hole parameters from gravitational-wave strain data, scaling training to **30,000+ simulations** with a CosineDecay learning-rate schedule over 250 epochs; used z-score contraction diagnostics to validate posterior calibration.

3D Point Cloud Reconstruction

Industrial Data Science Course

OpenCV, Open3D, SuperPoint, LightGlue

Reconstructed dense 3D point clouds from orbital video via **feature matching and triangulation**, using the Essential Matrix (5-DOF) formulation given known camera pose from orbital motion, generating **3.2M+ validated data points**.

Intercultural Competence Index

DiveHack 2026, TU Dortmund × Volkswagen Group

BERT, NLP, Python

Built a behaviour-based Intercultural Competence Index using **BERT-based NLP analysis** during a hackathon with Volkswagen Group.

titanic-odds

Introduction to Python – Final Project

Python

Building an installable **survival-prediction and model-explainability** Python package on the Titanic dataset, with a full test suite and documentation.

HONOURS & AWARDS

“AI-Based Interview System with Machine Learning” · Co-authored, Published in IJRPR Vol. 6, Issue 5 2025

Group project (5 co-authors) building an AI-driven interview simulator combining speech recognition, text-to-speech, and CNN-based facial emotion recognition with KNN-based resume screening; evaluated with 20 test participants at a 4.2/5 satisfaction rating.

“A Research-Based Project on Animal Species Detection” · Co-authored, Published in IJRPR Vol. 5, Issue 2 2024

Group project (5 co-authors) proposing a lightweight YOLOv2-based animal species classifier for wildlife-vehicle-collision prevention, evaluated across a 2,600+ image, 10-species dataset.

VOCATIONAL TRAINING

Python with Machine Learning · SEED IT Solution

Jul – Aug 2024

Bhilai, India

Completed vocational training in Python and machine learning fundamentals; graded A+.

Python with Data Science · Feasible Research & Technologies Pvt. Ltd.

Aug – Sep 2023

Raipur, India

Completed vocational training in Python and data science fundamentals under company mentorship.

LEADERSHIP & COMMUNITY

NSS Volunteer · Bhilai Institute of Technology

2021 – 2023

Raipur, India

Completed 240 hours of community service through the National Service Scheme (NSS), earning a ‘B’ grade certification.

LANGUAGES

Hindi, Native fluency **English**, Proficient (C1) **German**, A2, Actively learning, committed to reaching professional fluency